

Problem 2: Complete Solution

1. Build Hamiltonian and Mapping Comparison

To place a molecule on a quantum computer, one must first map the fermionic Hamiltonian to qubit Pauli operators. Jordan-Wigner (JW) and Parity are two common mappings — they are mutually exclusive alternatives. Closed-shell molecules possess N_α and N_β particle-number conservation symmetries ($\mathbb{Z}_2$ symmetries), each eliminating 1 qubit.

Given: Molecules (a) H_2 (STO-3G) and (d) N_2 (STO-3G).

Solve:

- (a) For H_2 , use PySCF to compute RHF and extract one-body integrals h_{pq} (p, q are spin-orbital indices) and two-body integrals $(pq|rs)$ (chemist's notation, p, q, r, s are spin-orbital indices). Write the second-quantized Hamiltonian $\hat{H} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} (pq||rs) \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r + E_{\text{core}}$.
- (b) Using $h_{01} \hat{a}_0^\dagger \hat{a}_1$ (subscripts 0,1 are spin-orbital indices) as an example, derive its Pauli string expansion under JW.
- (c) Count Pauli terms and maximum Pauli weight (non- I count) for H_2 under both mappings.
- (d) Use the two $\mathbb{Z}_2$ symmetries ($N_\alpha=1, N_\beta=1$) to reduce H_2 from 4 to 2 qubits. Write the reduced Hamiltonian explicitly; explain how JW and parity differ in achieving this.
- (e) Repeat (c) for N_2 (20 qubits).

Advanced Challenge: BK mapping has Z-string length $O(\log n)$, hence max weight far lower than JW. For N_2 ($n=10$), what is BK's max weight? How many times lower is the circuit depth?

Solution:

- (a)
- (b)
- (c)
- (d)
- (e)

2. HF State and SQD Subspace Diagonalization

SQD samples S bitstrings from a quantum state $|\Psi\rangle$, each corresponding to an electronic configuration (Slater determinant $|D_k\rangle$), and exactly diagonalizes the CI matrix $H_{kl} = \langle D_k | \hat{H} | D_l \rangle$ in the subspace $\mathcal{S} = \text{span}\{|D_k\rangle\}$. The simplest initial state is the HF state $|\text{HF}\rangle = \prod_{i \in \text{occ}} \hat{a}_i^\dagger |0\rangle$ — a single computational basis state.

Given: Molecule (a) H_2 , 4 spin-orbitals (indexed 0–3), 2 electrons.

Solve: Determine which computational basis state $|b_3 b_2 b_1 b_0\rangle$ ($b_i \in \{0, 1\}$) H_2 's HF state corresponds to under JW, and write the preparation circuit. Manually list all $\binom{4}{2}=6$ configurations, build the 6×6 CI matrix, exactly diagonalize, and compare with PySCF FCI energy. If the subspace contains only HF + all single excitations (no doubles), what does the SQD energy equal? Explain why using Slater-Condon rules and HF self-consistency.

Advanced Challenge: Stretch H_2 bond to 3.0 Å. Does the conclusion about HF + singles still hold? What happens to HF self-consistency in the strongly correlated regime?

Solution:

3. Configuration Recovery: Principle and Implementation

Quantum hardware noise causes sampled bitstrings to violate particle-number conservation. Configuration recovery uses the average occupancy $\bar{n}_i = \frac{1}{S} \sum_{s=1}^S b_i^{(s)}$ ($b_i^{(s)}$ = value of bit i in the s -th sampled bitstring) to correct erroneous bitstrings.

Given: H_2 's HF bitstring is $|0011\rangle$ (see Problem 2), noise flip probability 30%, sample count $S=1000$.

Solve:

- Generate $S=1000$ noisy bitstrings by independently flipping each bit with 30% probability. Implement configuration recovery: for each violating bitstring, flip the bit with largest $|b_i - \bar{n}_i|$ until constraints satisfied. Verify correctness.
- Derive: why is “flip the largest $|b_i - \bar{n}_i|$ ” optimal in the maximum likelihood sense?
- Increase noise rate from 10% to 50%. Plot recovery success rate vs. noise rate. Determine the failure threshold (50% success rate) and explain the physics.

Advanced Challenge: Formulate as Bayesian inference — $\bar{n}_i$ as prior, observed bitstring as likelihood, find MAP configuration. How does this relate to (b)?

Solution:

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4. LUCJ Quantum Circuit Gate Decomposition

Sampling from HF yields only 1 configuration. The LUCJ ansatz uses CCSD amplitudes to build a correlated state. Its UCJ operator is $\hat{U}_{\text{UCJ}} = \hat{U} e^{i\hat{J}} \hat{U}^\dagger$, where $\hat{U} = e^{\hat{\kappa}}$ is orbital rotation ($\hat{\kappa}$ anti-Hermitian), $\hat{J} = \sum_{i,j} J_{ij} \hat{n}_i \hat{n}_j$ is diagonal Coulomb. The “local” truncation retains only adjacent-qubit R_{ZZ} . Givens rotation = 2 CNOT + single-qubit gates; $R_{ZZ} = 2 \text{ CNOT} + 1 R_Z$.

Given: Molecule (a) H_2 (1 UCJ layer); $n=20$ qubits on a 1D chain.

Solve:

- Derive how $e^{iJ_{ij}\hat{n}_i\hat{n}_j}$ maps to R_{ZZ} . Hint: use $\hat{n}_i = \frac{I-Z_i}{2}$.
- Build the complete LUCJ circuit for H_2 , count CNOTs and depth.
- For $n=20$ on 1D chain, use edge coloring to argue: full UCJ R_{ZZ} depth $O(n^2)$, LUCJ $O(1)$. Give specific values.

Advanced Challenge: On 2D heavy-hex topology, what is LUCJ’s R_{ZZ} layer depth? Advantages over 1D?

Solution:

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5. LUCJ-SQD vs HF-SQD: How Entanglement Improves Sampling

The HF state is a computational basis state — measurement is deterministic, yielding only 1 configuration. The LUCJ state $|\Psi_{\text{LUCJ}}\rangle = \hat{U}_{\text{UCJ}}|\text{HF}\rangle$ contains CNOT gates creating entanglement, making measurement random and covering more configurations. Parameter `ccsd_scale` $\lambda \in [0, 1]$: $\lambda=0$ gives HF state.

Given: Molecule (b) LiH (STO-3G), $\lambda=0.05$, $S=1000$.

Solve:

- Sample $S=1000$ bitstrings from HF and LUCJ ($\lambda=0.05$) for LiH , feed into SQD, compare energy accuracy vs. FCI.
- Explain from quantum information perspective why LUCJ can cover more FCI configurations (key mechanism?), and discuss what $\lambda=0$ and $\lambda=1$ correspond to. Why use 0.05 instead of 1?

Advanced Challenge: Is there a quantitative relationship between LUCJ state’s entanglement entropy and required samples S ?

Solution:

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(b)

6. SQD vs VQE: Quantum Resources and Energy Accuracy (Open)

VQE prepares $|\Psi(\theta)\rangle$ (p parameters), computes gradients via parameter-shift ($2p$ evaluations per step), each energy evaluation measures $O(N^4)$ Pauli terms with $O(1/\epsilon^2)$ shots per term. SQD prepares a fixed state, samples S bitstrings, iterates N_{iter} times. Current NISQ: $p_{2q} \sim 0.01$.

Given: Molecule (b) LiH ($N=8$). VQE: $p=4$, $N_{\text{iter}}=100$; SQD: $S=1000$, $N_{\text{iter}}=5$; $\epsilon=10^{-3}$ Ha.

Solve:

- Quantum resources:* Compute total circuit executions for both, compare orders of magnitude.
- Energy accuracy:* Given $|\Psi\rangle = \sum_x c_x |x\rangle$, define $\mathcal{S} = \text{span}\{|x\rangle : c_x \neq 0\}$. Prove $E_{\text{SQD}} = \min_{|\phi\rangle \in \mathcal{S}} \langle \phi | \hat{H} | \phi \rangle \leq \langle \Psi | \hat{H} | \Psi \rangle = E_{\text{VQE}}$. When does equality hold?
- Balanced comparison:* List advantages/disadvantages of SQD and VQE in: circuit depth, mapping flexibility, ansatz choice, noise robustness, measurement overhead, variational scope. Which is more practical on NISQ ($p_{2q} \sim 0.01$)? In the FTQC era ($p_{2q} < 10^{-4}$)?

Advanced Challenge: Design a hybrid — VQE’s shallow circuit for state prep, SQD’s diagonalization for post-processing. What is the energy lower bound?

Solution:

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7. FCI Space Explosion and EWF’s Scaling Advantage

FCI dimension $\binom{2n}{N_e}$ grows exponentially. EWF splits into F fragments, each with n_{frag} orbitals and N_{frag} electrons, solved independently.

Given: Test molecules (a)–(e). C_2H_4 ’s EWF max fragment: 20 qubits ($n_{\text{frag}}=10$).

Solve:

- Compute FCI dimensions for all 5 molecules, tabulate, observe trend.
- Prove: fixed n_{frag} , $F = N/n_{\text{frag}} \rightarrow$ EWF cost $O(N)$.
- Compute C_2H_4 max fragment FCI dimension, compare with full $\sim 3.1 \times 10^7$.
- Derive crossover: direct FCI $O(\binom{2N}{N_e}^3)$ vs. EWF $F \times O(\binom{2n_{\text{frag}}}{N_{\text{frag}}}^3)$. At what n_{frag} are they equal?

Advanced Challenge: Under 20-qubit NISQ constraint ($n_{\text{frag}} \leq 10$), optimal n_{frag} for N_2 and C_2H_4 ?

Solution:

- (a) 直接计算即可, H_2 是 $\binom{4}{2} = 6$, LiH 是 $\binom{8}{4} = 70$, H_2O 是 $\binom{14}{10} = 1001$, N_2 是 $\binom{20}{14} = 38760$, C_2H_4 是 $\binom{28}{16} = 30421755$ 。
- (b) 当 n_{frag} 固定时, 每个片段的 FCI 维数 $\binom{2n_{\text{frag}}}{n_{\text{frag}}}$ 有上界, 其对角化代价为 $O(D_{\text{frag}}^3) = O(1)$, 片段总数为 $F = N/n_{\text{frag}}$, 所以 EWF 总代价为 $O(N)$ 。
- (c) C_2H_4 最大碎片为 20 量子比特, 该碎片的 FCI 维数最大是 $\binom{20}{10} = 184756$, 全分子的 FCI 维数最大是 30421755, 二者之比

$$\frac{184756}{30421755} \approx 0.6\%.$$

- (d) 假设电子均匀地分布在每个碎片中, 得到方程

$$\binom{2n_{\text{frag}}}{N_e n_{\text{frag}}/N} = \binom{2N}{N_e} \left(\frac{n_{\text{frag}}}{N}\right)^{1/3}.$$

我们来证明: 对于 $n_{\text{frag}} < N$, 左边总是比右边小很多倍, 进而 EWF 总是比直接的 FCI 需要更少的时间复杂度 (显然当 $n_{\text{frag}} = N$ 时是相等的)。

设 $\frac{N}{n_{\text{frag}}} = t$, 把左边的组合数记为 $\binom{a}{b}$, 有如下推导:

$$\binom{2N}{N_e} = \binom{ta}{tb} = \prod_{i=1}^{tb} \frac{ta-i+1}{tb-i+1} = \prod_{\substack{i=1 \\ t \nmid i-1}}^n \frac{ta-i+1}{tb-i+1} \cdot \binom{a}{b} \geq (ta-tb+1) \binom{a}{b} > \sqrt[3]{t} \binom{a}{b}.$$

证明完毕。

8. DMET Fragmentation: Schmidt Decomposition and Bath Exactness

EWF uses Schmidt decomposition: $|\Psi\rangle = \sum_{k=1}^r \lambda_k |u_k\rangle \otimes |v_k\rangle$ ($\lambda_k \geq 0$ Schmidt coefficients, r Schmidt rank), retaining n_{bath} largest- λ_k environment states as bath. DMET minimal bath ($n_{\text{bath}}=n_{\text{imp}}$) is exact at HF level (γ^{HF} is a projector: $\gamma^2=\gamma$), but becomes inexact when solver is upgraded.

Given: Molecule (c) H_2O , DMET atomic fragmentation.

Solve:

- (a) Prove $r \leq \min(\dim \mathcal{H}_{\text{imp}}, \dim \mathcal{H}_{\text{env}})$, derive bath upper bound n_{imp} .
- (b) Prove DMET minimal bath exactly reproduces HF density matrix. Key: projector property.
- (c) When solver upgrades to CCSD/SQD, $\gamma^{\text{corr}} \neq (\gamma^{\text{corr}})^2$. Why is minimal bath no longer exact? What correlation does MP2 bath supplement?

Advanced Challenge: How is DMET's self-consistent μ determined? Oneshot vs. self-consistent DMET trade-offs?

Solution:

- (a)
- (b)
- (c)

9. Energy Reconstruction: Algebraic Elimination of Double-Counting

Naive sum $\sum_f E_f$ is too large (overlapping bath HF contributions counted twice). Standard reconstruction: $E_{\text{total}} = E_{\text{MF}} + \sum_f (E_f - E_{\text{MF},f})$. Fragment energy decomposes: $E_f = E_{\text{core},f} + E_{\text{elec},f}$ (core = nuclear repulsion + frozen core, solver-independent; electronic = solver-dependent).

Given: Molecule (c) H_2O , fragments O and H_2 . Values (Ha): $E_{\text{MF}} = -74.963$, $E_{\text{O}} = -75.010$, $E_{\text{MF,O}} = -74.980$, $E_{\text{H}_2} = -75.005$, $E_{\text{MF,H}_2} = -74.983$.

Solve: Substitute $E_f = E_{\text{core},f} + E_{\text{elec},f}$ and $E_{\text{MF},f} = E_{\text{core},f} + E_{\text{HF-elec},f}$ into $(E_f - E_{\text{MF},f})$. Prove $E_{\text{core},f}$ cancels, leaving $\Delta E_{\text{corr},f} = E_{\text{elec},f} - E_{\text{HF-elec},f}$. Why is $\Delta E_{\text{corr},f}$ local and free from double-counting? Verify: HF fragments $\rightarrow E_{\text{total}} = E_{\text{MF}}$. Compute reconstructed total energy.

Advanced Challenge: Partitioned cumulant reconstruction — basic idea? When significantly more accurate?

Solution:

10. Full Pipeline Integration and End-to-End Error Analysis (Open)

Chaining Problems 1–9 gives the complete EWF-SQD pipeline: RHF $\rightarrow$ DMET fragmentation $\rightarrow$ per-fragment LUCJ-SQD $\rightarrow$ energy reconstruction. Three error sources: ϵ_{frag} (bath truncation, decreases with larger fragments), ϵ_{SQD} (incomplete sampling, decreases with more samples S), ϵ_{noise} (quantum noise, related to circuit depth and p_{2q}).

Given: Molecules (a) H_2 , (c) H_2O , (d) N_2 , (e) C_2H_4 . Fixed budget B . Error model $\epsilon_{\text{total}} \approx \epsilon_{\text{frag}} + \epsilon_{\text{SQD}}$, cost model $B = c_1 N n_{\text{frag}}^3 + c_2 N S^3 / n_{\text{frag}}$.

Solve:

- (a) Run full pipeline on all 4 molecules, report E_{MF} , $E_{\text{EWF-SQD}}$, E_{FCI} (or CCSD).
- (b) Under fixed B , use Lagrange multipliers to derive optimal allocation $\epsilon_{\text{frag}} \approx \epsilon_{\text{SQD}}$ (“barrel effect”).
- (c) Discuss: H_2O (14q) $\rightarrow$ C_2H_4 (28q), which error grows fastest? N_2 bond stretched to 2.0 Å, which degrades fastest?

Advanced Challenge: In FTQC era, SQD $\rightarrow$ QPE. How does EWF reduce QPE resources from $O(N/\epsilon)$ to $O(n_{\text{frag}}/\epsilon)$? At what p_{2q} does QPE outperform SQD?

Solution:

- (a)

(b)

(c)